H/E/TZ

Reg. No: 23BCE2093

Final Assessment Test – April 2026

VIT® Vellore Institute of Technology

Course: **BCSE305L - Embedded Systems**

Class NBR(s): **1643/1644/1648/1655/1659/1663/1667/1671/1673/1676/1680/1688/1690/1694/1698/5593/5872** Slot: **G1+TG1**

Time: **Three Hours** Max. Marks: **100**

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

COs	CO Statements
CO1	Identify the challenges in designing an embedded system using various microcontrollers and interfaces.
CO2	To summarise the functionality of any special purpose computing system and to propose smart solutions to engineering challenges at the prototype level.
CO3	To examine the working principle and interface of typical embedded system components, create programme models, apply various optimization approaches including simulation environment and demonstration using debugging tools.
CO4	To evaluate the working principle of serial communication protocols and their proper use as well as to analyze the benefits and drawbacks of real-time scheduling algorithms and to recommend acceptable solutions for specific challenges.

BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions
(10 X 10 = 100 Marks)

1.	a) PIC-16 microcontrollers are much preferred for industrial purposes. With a neat block diagram, pinout diagram explain the functioning of a PIC-16 microcontroller highlighting the features that support automation.	[7]	CO1	BL3
	b) What is role of oscillator in PIC-16?	[3]		
2.	Discuss the mode 1 operation of the timer in 8051 and elaborate on how it can be configured as a watchdog timer. Explain role of watch dog timer with a case study.		CO1	BL4
3.	The speed of DC motor should vary based on the temperature of the environment. Choose appropriate ADC, sensors and actuators and design the same with a block diagram and a pseudocode for a case study.		CO1	BL4
4.	Existing types of attendance capturing devices in the classrooms have many challenges. Discuss the challenges and suggest an image capturing system which ensures that a student was physically and mentally present in the classroom for the entire period. Design the system with the Hardware requirement specification (HRS) and the system diagram.		CO2	BL3

5. Optimize the given code and discuss how code optimization is beneficial in embedded system design. CO3 BL3

```
void calculate(int *arr, int n, int factor) {
  for (int i = 0; i < n; i++) {
      arr[i] = arr[i] + (factor * 2) + (n + 5);
    int extra = (n + 5) * 2;
    int a=10/2;
  }
}
```

6. In a factory, the gas monitoring operation takes 2ms and is repeated every 5 ms, temperature analysis takes 3 ms and is repeated every 7 ms, alarm logging takes 4 ms and is repeated every 10 ms. Analyse the given operations with a time line chart for 20 ms and find which deadline is missed when EDF scheduling is applied. Suggest any method to avoid this issue. CO4 BL4

7. A deadline is missed in the given scenarios. CO4 BL4

Scenario 1: Pacemaker in a patient
Scenario 2: Industrial robot in manufacturing
Scenario 3:Online video streaming platform
Scenario 4:Navigation app with traffic updates

Identify whether the systems are hard/soft real time system. Justify the consequence of missing the deadline and suggest means to overcome the same.

8. List the specifications of RS232, Wifi, Zigbee and Bluetooth with a real time case study for each hardware module. CO4 BL3

9.a) A pick and place robot is used to sort 5 different types of fruits and place them in different baskets. The number of fruits in each basket is displayed on the display. Design an embedded system to implement the Robotic operation by giving the Hardware requirement specification, block diagram, software flow diagram and code. CO2 BL3

OR

9.b) There is a need to identify the pain experienced by patients in ICU, apart from their blood pressure, temperature and oxygen levels. An alert signal is sent to the nurses when anyone of the levels crosses the threshold. Design an embedded system to implement with appropriate Hardware requirement specification, block diagram, software flow diagram and code. CO2 BL3

10.a) Cyber security is of much importance in the Industrial revolution 5.0. List down any 5 security concerns in implementing embedded systems in the healthcare system. CO2 BL2

OR

10.b) Mobiles are continuously evolving, with devices becoming smaller while offering more advanced functionalities. Identify five key features and the corresponding technologies that have contributed to the improvement and advancement of each mobile model. CO2 BL2

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